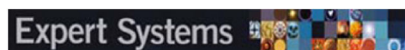


## SPECIAL ISSUE PAPER



WILEY

# Modelling material flow using the Milk run and Kanban systems in the automotive industry

Dragan Simić<sup>1</sup> | Vasa Svirčević<sup>2</sup> | Emilio Corchado<sup>3</sup> | José L. Calvo-Rolle<sup>4</sup> | Svetislav D. Simić<sup>1</sup> | Svetlana Simić<sup>5</sup>

<sup>1</sup>Faculty of Technical Sciences, University of Novi Sad, Novi Sad, Serbia

<sup>2</sup>Lear D.O.O, Novi Sad, Serbia

<sup>3</sup>Departamento de Informática y Automática, Universidad de Salamanca, Salamanca, Spain

<sup>4</sup>Department of Industrial Engineering, University of A Coruña, A Coruña, Spain

<sup>5</sup>Faculty of Medicine, University of Novi Sad, Novi Sad, Serbia

## Correspondence

Dragan Simić, Faculty of Technical Sciences, University of Novi Sad, Trg Dositeja Obradovića 6, 21000 Novi Sad, Serbia.  
Email: dsimic@eunet.rs

## Abstract

Material flow management refers to the analysis and specific optimization of the inventory-production system. Material flow can be characterized as the organized flow of material in a production process with the required sequence determined by a technological procedure. The *Milk run* system assures the transportation of materials at the right time and in an optimal manner. It should be combined with the Kanban system to highlight when something is required in the production process. This paper presents biological swarm intelligence, in general, and a particular model, particle swarm optimization (PSO), for modelling material flow using a *Milk run* system supported by a Kanban system in the automotive industry. The aim of this study is to create a new model for the optimal number of trailers for one train and optimal number of containers in a tugger train system when the route time period has been defined. A new modified PSO approach for integrating inventory-production in a unique optimization model is used. The major modification to the original PSO is using the capacity of a container instead of a velocity component. Each new Kanban trigger is checked, and the total timing for the *Milk run* delivery solution is calculated for the necessary raw material capacity for each shop floor.

## KEYWORDS

Kanban, material flow, Milk run, particle swarm optimization

## 1 | INTRODUCTION

Material flow management refers to the analysis and specific optimization of material and energy flows that arise during the manufacturing of products and provision of services. Material flow can be characterized as an organized flow of material in a production process with the required sequence determined by a technological procedure. It is a summary of operations presented by material conveying, storage, packaging and weighing, and technological manipulations and other works directly related to the production process. Planning and dimensioning material flow challenges are difficult to overcome, especially in scenarios characterized by many hard constraints and by well-established processes (Wagner, 2006).

Furthermore, we should differentiate between internal and cross-company material flow management; whereas internal material flow refers to the movements of substances or materials within a company (Simić, 2012), cross-company (external) material flow describes the path of a material along the value-added chain (Simić, 2015). Thus, successful material flow management links the structural analysis of material flow with data available in business information systems with recent paradigms, such as enterprise resource planning, logistics information systems, big data, artificial intelligence, in general, and soft-computing techniques, in particular (Ilin, 2017).

A *Milk run* system is one of the most efficient and popular models used to improve logistic operations in every serial production environment such as the automotive industry, electrical industry, electronic industry, mechanical industry, food distribution, and military. It can effectively reduce inventory, while increasing the material flow between warehouses and shop floors, and the usage rate of the vehicles, therefore decreasing costs and improving productivity (Simić-1, 2019a, 2019b). The use of a *Milk run* system is based on tugger trains for industrial material transportation, which have shown three important tendencies: (a) tugger train systems are being mainly adopted for material supply; (b) adoption of tugger train systems has recently increased following a linear trend and (c) the use of tugger train systems in other sectors is consistently growing (Urdu, 2018).

However, a *Milk run* system has some limitations. It must be implemented with other material flow systems, such as a traditional Kanban system, an *E-kanban* system, some sets of *constant work-in-progress* (CONWIP) mechanisms, or software for material needs control. Therefore, this study focuses on systems that incorporate the existing *Milk run* production-delivery system presented in (Simić-2, 2019a, 2019b), which regulates both the Kanban system and complete material flow in an inventory-production system. The *Milk run* system assures that inventory is transported at the right time and in an optimal way. The objective of this paper is to present the inception of the pull system and the analyses of system performance based on simulation study.

This paper presents biological swarm intelligence, in general, and a particular model, the modified particle swarm optimization (PSO), for modelling material flow using a *Milk run* system supported by a Kanban system for use in the automotive industry. The aim of this research is to create model to minimize inventory-production costs.

The rest of the paper is organized as follows: Section 2 provides an overview of material flow management and a review of related studies. Section 3 describes the PSO implemented in material flow using the *Milk run* system and Kanban system. This section also describes the dataset used in the implementation. Experimental results and a discussion are presented in Section 4, and finally, Section 5 gives the concluding remarks.

## 2 | MATERIAL FLOW AND RELATED WORK

The term material flow covers a broad spectrum of methods and approaches in the literature. Most contemporary organizations are facing many dilemmas, the most important being determination of ways of acquiring or maintaining their position in a competitive environment. Endeavours to reduce inventory, in the context of operation management, are often related to lean manufacturing, which focuses on removing waste from activities and processes that do not add value to products, thus creating an efficient inventory-production system (Chen, 2010).

The strategic aim of all production enterprises is to achieve the highest profit possible. Toward this end, attributes of lean production that are related to and support the reduction of production costs are used. By using theoretical knowledge and a detailed analysis of a material flow, including mapping of individual production processes, it is possible to obtain the necessary data sets that identify savings options by increasing efficiency, including cost savings. The methods for modelling and simulation used for the optimization of material flow are presented in (Kodym, 2019).

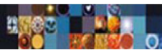
Competitive prevalence can be achieved by analysing and optimizing material flow, which contributes to a reduction of the company costs. Material flow in a production organization shows how the production system is organized. In research paper (Krolczyk, 2015), the current state of material flow in a production company in the automotive sector is analysed. Material flow optimization has been proposed, and it consists of the creation of a program of internal transport and optimization of the location of the working stands on the shop floors. Furthermore, the current problems of the company's internal transport system are highlighted and solutions to those problems are proposed.

Material handling of flow is one of the most crucial issues that should be considered to eliminate waste and reduce the cost and delivery of the product based on the "just in time" method. An effective transportation management system optimized with the *Milk run* concept, must be implemented to control the cost of transportation and inventory. The *Milk run* material supply system consists of cyclic trips, where goods are collected from one supplier and delivered to several customers. The objective of this optimization is minimization of the total material handling and inventory holding costs, which could boost sales or increase the profit margin of the organization (Patel, 2017).

A systematic approach which consists of *Milk run* route optimization with a mixed integer linear programming model that comprises a container loading heuristic that increases the utilization of *Milk run* vehicles and a time interval assignment heuristic that prevents overlapping of *Milk run* vehicles at the depots of a manufacturer has been proposed in (Aksoy, 2016).

A mixed-integer nonlinear programming (MINLP) model to determine the number of Kanban and the transported batch size in a multistage supply chain system was developed in (Sarker, 1998). In this regard, the objective was to minimize inventory holding, transportation of each container, and setup costs.

System metaheuristic methods are used to determine the optimal Kanban quantity and withdrawal lot sizes in a supply chain. A genetic algorithm and a hybrid of a genetic algorithm and simulated annealing (SA) are used (Widyadana, 2010). To reduce the computational time required to obtain an optimal solution, PSO and SA algorithms as metaheuristic methods are used to solve a large MINLP to determine the number of Kanban in a cellular manufacturing system with consideration of the rework process (Aghajani, 2012).



A case study and simulation analysis on a Kanban system incorporating a *Milk run* operation used to add raw materials to the production process is presented in (Chee, 2012). Here, the study data were collected for 10 weeks. The proposed Kanban model was simulated to test its various performances including total output, average flow time, average work-in-process, and average waiting time. Response surface methodology was adopted to generate suitable representative regression models. For all performance measures, the simulation results revealed that the proposed system consistently outperforms the push system currently used in the company. The operators in the proposed Kanban system have a lower utilization, even with the additional material handling task.

The problems of transportation planning have characteristics that match the specific skills of multi agent systems (MAS). These systems are able to cope with inter-organizational and event-oriented planning. The study presented in (Pinheiro, 2019) evaluates different collaboration strategies for dynamic vehicle routing and autonomous vehicle interaction as well as communication between vehicles in *Milk run* operations using MAS, which allow vehicle interaction and autonomous responses in unplanned events. Auxiliary vehicles are used to meet the tasks that cannot be met by regular vehicles with unnecessary additional cost.

In the logistics system, there are many assembly plants and numerous part suppliers located in close proximity to each other. The suppliers collect the parts and transport them to the assembly plants in the small-batch and multi-frequency mode. A multi-factory *Milk run* pickup problem with uncertain demand and frequency in China's automobile industry is presented in (Wu, 2018). Considering the balance between inventory cost and distribution cost, a mixed-integer programming model was built for this problem and converted into a robust optimization model by the Chernoff–Hoeffding theorem. Then the adaptive genetic algorithm and local search were combined into a general hybrid heuristic algorithm to successfully solve the problem.

The fundamental inventory-production problem that needs to be solved is the problem of bringing each component to the line only at the moment and at the quantity needed to feed the assembly line. The introduction of decentralized storage areas, called supermarkets (SM) made it possible for frequent deliveries of small batches of components to be made (Emde, 2017). Supermarkets are located between the production line and the central warehouse. A discrete event simulation model is developed, and a mathematical model based on a PSO optimization is used to support the simulation. The hybrid model is finally applied to practical situations and presented in (Kundua, 2019).

The practical implementation of a *Milk run* system by Toyota Group in automotive factories in Indonesia was analysed in (Hardi, 2019). The *Milk run* system was designed to collect goods from two or more suppliers and send them to the customer. At the same time, the capacity of a truck was optimized to increase the cargo for fully loaded distribution. The time and number of pickup schedules of the *Milk run* determine the performance of the logistics system, while the route design and scheduling are the key factors that have an impact on the cost and also determine the smoothness of the continuous pickup. The key goal of a *Milk run* operation is to minimize the inventory (holding) cost and transportation cost. The advantage of a *Milk run* operation is cost reduction, but it also supports green logistics by decreasing carbon dioxide emissions (CO<sub>2</sub>) by reducing the number of trucks used.

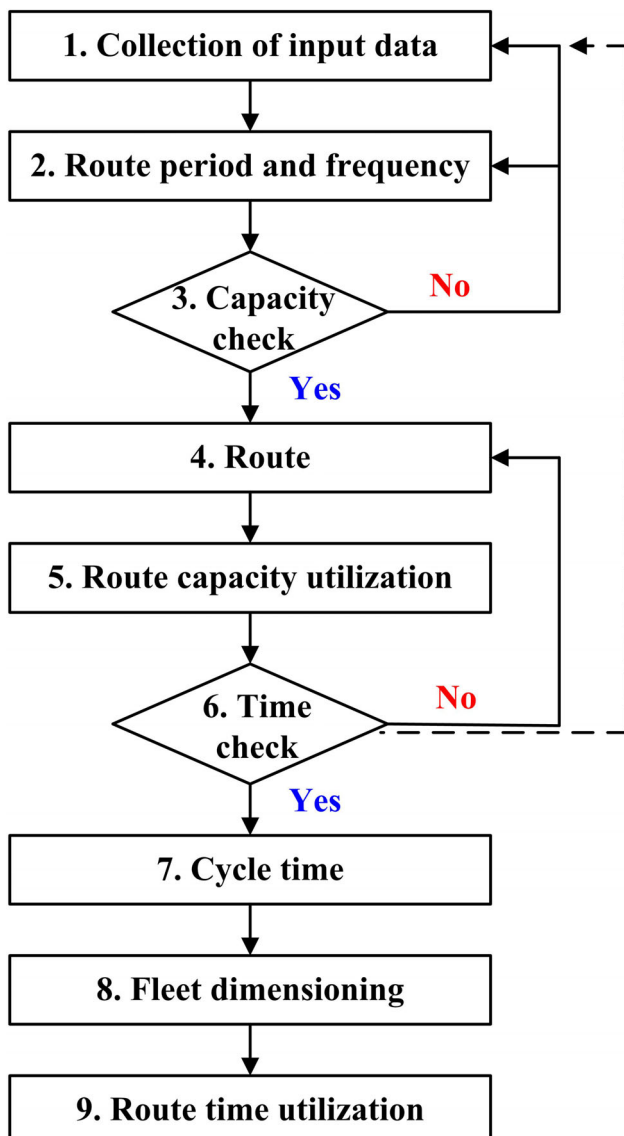
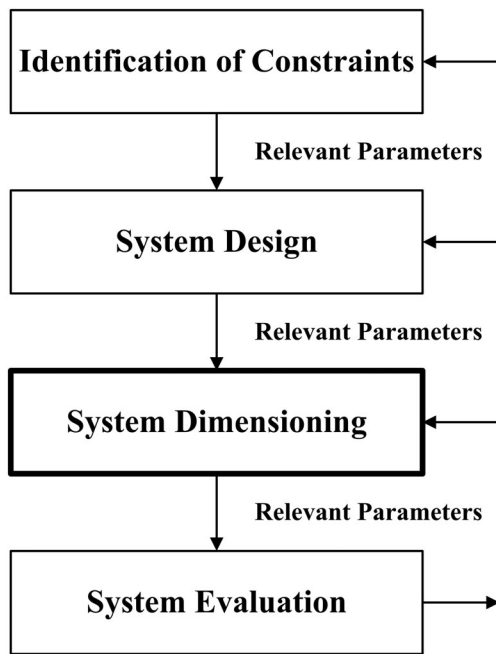
An optimization model implemented in a hybrid environment that contains constraint logic programming, mathematical programming, and metaheuristics for the capacitated vehicle routing problem with pick-up and alternative delivery is presented in (Sitek, 2019). The zero-inventory approach, which has been widely adopted by many automobile companies, seems to be impractical in certain real production contexts as it requires an inventory of all parts but in low amounts. A mixed integer programming formulation for a new model that can simultaneously determine the trip routes for collecting automobile parts and minimize total production costs and inbound logistics is shown in (Mao, 2019).

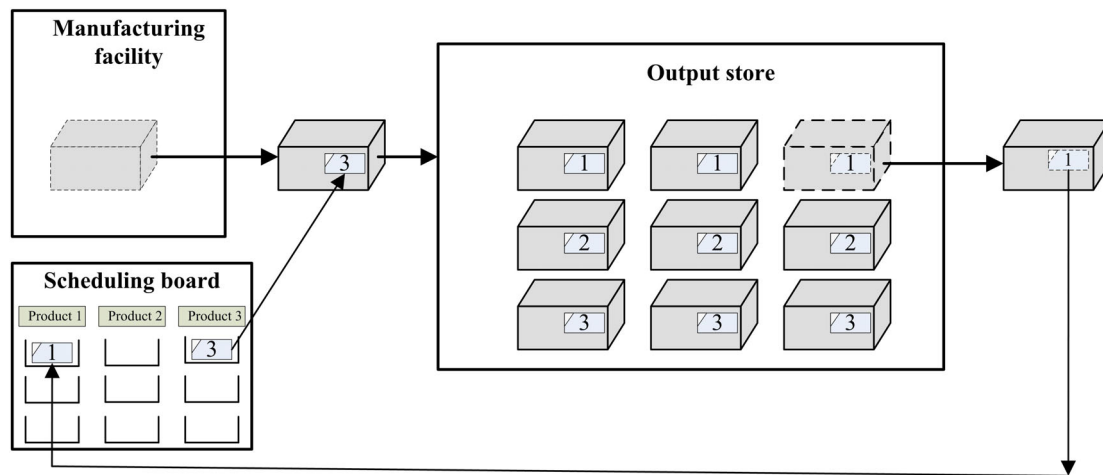
### 3 | MODELLING THE MILK RUN AND KANBAN SYSTEM IN MATERIAL FLOW

To obtain a perfect inventory-production system based on the *Milk run system*, most of the operations should be automated and controlled. Over the years, many production control systems have been proposed within various factories, such as push-like buffered production lines, pull-like Kanban systems, and hybrid-like CONWIP (Song, 2013). For buffered production lines the input quantities are the sizes of the buffers between each of the machines, that is, the workplaces. For Kanban systems, the input quantities are the number of production and transfer Kanbans. For CONWIP systems, there is only one input quantity, which is the number of CONWIP cards (Herer, 2000).

#### 3.1 | Milk run system

The basic planning process of a *Milk run system* is divided in four phases, as is shown in Figure 1. Every choice made in one phase impacts the next. This impact is not easily assessable until all the planning has been completed. Therefore, the planning process is based on a loop system so that, in every planning phase, decisions made in the previous phases can be called into question, and relevant parameters may be adjusted. This planning approach requires a high level of expertise and competence to ensure that the right adjustments are made and a consistent and useful result is obtained.

**FIGURE 1** Planning phases of a Milk run system**FIGURE 2** Steps of the dimensioning system of a Milk run system



**FIGURE 3** Basic Kanban system with three different products (Krieg, 2005)

The system dimensioning phase will be tackled here, and a new method will be introduced (Figure 2.). It will, therefore, be possible to follow, from the early stage of the dimensioning phase, both the material-flow and the information-flow, together with other requirements that characterize a typical scenario in enterprises where freedom of logistic planner is often restricted.

Accordingly, the development of a homogeneous perception of material flows and information flows is one of the central tasks of material flow management.

### 3.2 | Kanban system

Kanban systems were initially developed in Japan to manage multistage manufacturing systems. Every manufacturing stage has a target local inventory level for the local inventories, where the local inventory can be considered the inventory space at one shop floor. The customer demand information is transmitted upward stage by stage.

More specifically, it states that, whenever an item is withdrawn from the local inventory, a card is issued to the corresponding production unit to signal that a new part should be produced (Duri, 2000). This policy also operates on a pull mechanism. Its advantage is that the number of parts at each stage is limited by the number of Kanbans at that stage. The disadvantages are the indirect flow of demand information from customers to suppliers and the lack of utilization of the inventory information at downstream stages.

A CONWIP system can be viewed as a single stage Kanban system, in which a single type of card is used to control the total amount of inventories permitted in the entire system. Essentially, only one target inventory level for the entire system is triggered when the most upstream production unit needs to produce a new part (Takahashi, 2005).

Traditionally, a Kanban is a tag-like card (*Kanban* is Japanese meaning “a card” or “visible record”). One Kanban card must be attached to each container in the output store. The number of Kanbans is limited, restricting the maximum number of finished items in the system. When a container is withdrawn, the accompanying Kanban is detached from the container and placed on the scheduling board. Alternatively, the Kanban may be detached when the last item is removed from the container or box. Figure 3 shows the basic Kanban system with three different products (Krieg, 2005).

### 3.3 | PSO algorithm

The original PSO algorithm was inspired by the social behaviour of biological organisms, specifically the ability of groups of some species of animals to work, as one, in locating desirable positions in a particular area, for example, birds flocking to a food source.

PSO is originally attributed to Kennedy and Eberhart (Kennedy, 1995). The PSO algorithm is comprised of a collection of particles that move around the search space. An individual particle  $i$  is composed of three vectors: its position in the  $N$ -dimensional search space of solutions  $\vec{x}_i = (x_{i1}, x_{i2}, \dots, x_{iN})$ , the best position that it has individually found  $\vec{p}_i = (p_{i1}, p_{i2}, \dots, p_{iN})$ , and its velocity  $\vec{v}_i = (v_{i1}, v_{i2}, \dots, v_{iN})$ . Particles were originally initialized in a uniform random manner throughout the search space; the velocity was also randomly initialized. These are also weighted matrixes  $r_p$ ,  $r_g$  dimensions—search space  $\times$  time horizon—where items are random numbers of a unimodal distribution in the range  $U < 0, 1 >$ . These collections of

particles then move throughout the search space influenced by their own best past location and the best past location of the entire swarm or a close neighbour.

The Euclidian neighbourhood model was abandoned in favour of less computationally intensive models for mathematical optimization. Each iteration particle's velocity is updated using.

$$v_i(k+1) \leftarrow \omega v_i(k) + c_1 r_p (b_i - x_i(k)) + c_2 r_g (h_i(k) - x_i(k)) \quad (1)$$

where  $\omega = 1$  is the inertia weight;  $c_1 = 2$  is the personal learning coefficient;  $c_2 = 2$  is global learning coefficient;  $v_i(k+1)$  is the new velocity for the  $i$ th particle;  $x_i(k)$  is the  $i$ th particle's position at time  $k$ ;  $b_i$  is the  $i$ th particle's best-known position; and  $h_i(k)$  is the best position known to the swarm. Furthermore, a random matrix  $r_p(k) \sim U < 0, 1 >$  for  $k \in [1, N]$  and another random matrix  $r_g(k) \sim U < 0, 1 >$  for  $k \in [1, N]$  are generated. A particle's position is updated using  $x_i(k+1) \leftarrow x_i(k) + v_i(k)$ . In a PSO cycle, this process is repeated for all particles (individuals) in all the next generations. Algorithm 1 presents the PSO algorithm used in this study, where the entire swarm is updated at each step by updating the velocity and position of each particle in every dimension when the termination criteria is not satisfied or *MaxIt* maximal number of iterations is not satisfied.

---

**Algorithm 1** *The Particle Swarm Optimization algorithm*


---

**Begin**

- Step 1: Initialization.** A random population of individuals  $\{x_{ij}, i \in [1, N]$   
 Each individual's  $i$ -element velocity vector  $\{v_{ij}, i \in [1, N]$   
 The best-so-far position of each individuals:  $b_i \leftarrow x_i, i \in [1, N]$   
 Neighbourhood size  $\sigma < N$ ;  $nPop = 150$ ; Pop. Size; Max. velocity,  $v_{max}$ ;  
*MaxIt*=1000; Max. Num. of Iter.  $w=1$ ; Inertia weight  
 $c_1=2$ ; Personal Learning Coeff.;  $c_2=2$ ; Global Learning Coeff.
- Step 2: While** (the termination criteria are not satisfied) **or** (*MaxIt* = 1000)
- Step 3: for each** individual  $x_i, i \in [1, N]$   
      $H_i \leftarrow \{\sigma \text{ nearest neighbors of } x_i\}$   
      $h_i \leftarrow \arg \min_x \{f(x) : x \in H_i\}$   
     Generate a random matrix  $r_p(k) \sim U < 0, 1 >$  for  $k \in [1, N]$   
     Generate a random matrix  $r_g(k) \sim U < 0, 1 >$  for  $k \in [1, N]$   
      $v_i(k) \leftarrow \omega v_i(k) + c_1 r_p (b_i - x_i(k)) + c_2 r_g (h_i(k) - x_i(k))$   
     **if**  $|v_i(k)| > v_{max}$  **then**  
          $v_i(k) \leftarrow v_i(k) v_{max} / |v_i(k)|$   
     **end if**  
      $x_i(k) \leftarrow x_i(k) + v_i(k)$   
      $b_i \leftarrow \arg \min \{f(x_i(k)), f(b_i)\}$   
     **end for**  $i$  Next individual
- Step 4: end while** Next generation
- Step 5: Post-processing the results and visualization;**

**End.**

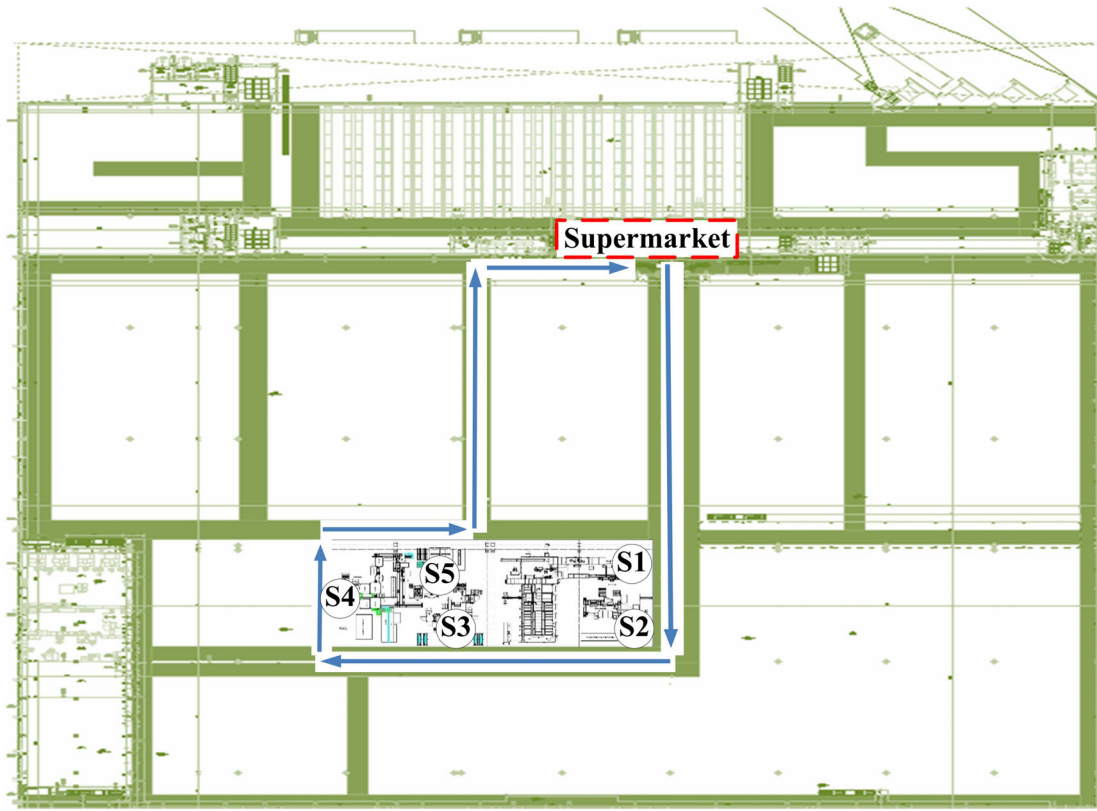
---

### 3.4 | Modified particle swarm optimization algorithm for the Milk run and Kanban systems

Some modifications of the PSO algorithm have been developed for tugger train planning using *Milk run* and Kanban systems. Daily delivery of products from the SM to the Kanban shop floor shelves, in a warehouse in Kanban locations where workers place the products according to the *plan for every part* (PFEP) in their proper positions, hourly, or according to the production plan.

The major aim of the optimization is to deliver the missing parts from the SM warehouse to the shop floor (assembly location) before the parts are needed. As the modified PSO Algorithm is a continuous solution for the problems of dynamic routing of the tugger train transport capacities and those generated during the inventory-production operations according to the priority of the parts needed, the major modification to the original PSO is use of the capacity of a container instead of a velocity component, as the tugger train transport operations and capacities are on





**FIGURE 4** Inventory-production system layout and space constraints

predefined paths. At each step, a new *trigger* is checked, and the total timing for the delivery solution is calculated for the necessary raw material capacity of the assembly line:

$$f(C_{t+1}^j) = w_t f(C_t^j) + c_1 r_1 (p_t^j - x_t^j) + c_2 r_2 (p_t^q - x_t^j) \quad (2)$$

where  $f(C_t^j)$  is the target capacity function as indexed;  $p_t^q$  is the global minimum capacity with the selected target among the farthest target;  $j$ th is the Kanban shop floor;  $p_t^j$  is *Kanban product level* at current time  $t$  in position  $i$ .

$$cap_{toGo}^j = \frac{X_{toWarehouse} + X_{toAssemblyshop}}{\text{raw material replenishment frequency}} \quad (3)$$

$$cap_{toProduction}^p = \text{Abs}(A_{missing}^p - A_{production}^k) \times q_k \quad (4)$$

where  $A_{missing}^p$  is the assembly number to which the missing part  $p$  belongs;  $A_{production}^k$  is the current assembly not within the assembly location  $k$ ; and  $q_k$  is time, which is defined as  $(1/\text{raw material replenishment frequency})$ .

To optimize the inventory-production system, the main objective is to minimize the total available capacity.

$$\text{minimize} \sum_{i=1}^{\# \text{ of trigger}} cap_{toProduction}^p - cap_{toGo}^j \quad (5)$$

Finally, the most important constraint is  $\forall p \in \{\text{trigger}\} cap_{toGo}^j \leq cap_{toProduction}^p$ , which provides, on one hand, the minimum capacity in tugger train transport, and on the other, the minimum capacity on the shop floor, and a smooth inventory-production process at the same time.

**Algorithm 2** The Modified Particle Swarm Optimization algorithm**Begin**

**Step 1: Initialization.** A random population of individuals  $\{x^j\}$ ,  $j \in [1, N]$   
 Each individual's  $j$ -element capacity vector  $\{C^j\}$ ,  $j \in [1, N]$   
 The best capacity of each individuals:  $cap^j \leftarrow x^j$ ,  $j \in [1, N]$   
 Neighbourhood size  $\sigma < N$ ;  $nPop = 150$ ; Pop. Size; Max. capacity,  $cap_{max}$ ;  $MaxIt = 1000$ ; Max. Num. of Iter.  $w = 1$ ; Inertia weight  
 $c1 = 2$ ; Personal Learning Coeff.;  $c2 = 2$ ; Global Learning Coeff.

**Step 2: While** (the termination criteria is not satisfied) **or** ( $MaxIt = 1000$ )

**Step 3: for** each individual  $x^j$ ,  $j \in [1, N]$

$H^j \leftarrow \{\sigma \text{ nearest neighbors of } x^j\}$

$H^j \leftarrow \{\sigma \text{ nearest neighbors of } x^j\}$

$h^j \leftarrow \arg \min_x \{f(x) : x \in H^j\}$

Generate a random matrix  $r_1(k) \sim U(0, 1)$  for  $j \in [1, N]$

Generate a random matrix  $r_2(k) \sim U(0, 1)$  for  $j \in [1, N]$

$f(C_{t+1}^j) = w_t f(C_t^j) + c_1 r_1(p_t^j - x_t^j) + c_2 r_2(p_t^j - x_t^j)$

**if**  $|C_{t+1}^j| > cap_{max}$  **then**

$C_{t+1}^j \leftarrow C_{t+1}^j cap_{max} / |C_t^j|$

**end if**

$cap_{toGo}^j = \frac{X_{toWarehouse} + X_{toAssemblyshop}}{\text{raw material replenishment frequency}}$

$cap_{toProduction}^p = Abs(A_{missing}^p - A_{production}^k) \times q_k$

minimize  $\sum_{i=1}^{\# \text{ of trigger}} cap_{toProduction}^p - cap_{toGo}^j$

**end for**  $j$  Next individual

**Step 4: end while** Next generation

**Step 5: Post-processing the results and visualization;**

**End.**

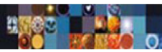
### 3.5 | Collection of input data

The first step of the method is collection of input data. Generally, the input data are based on the decisions the planner has made in the identification of constraints and system design phases, such as the data concern *layout* and *space constraints* of the inventory-production system, and the number of buffer places available at each stop (shop floors: S1, S2, S3, S4, and S5), which is presented in Figure 4. The *Milk run area* is often called the SM. It is located between the production line and the central warehouse and allows frequent deliveries of small batches of components.

This set of data is fundamental because it has a direct influence on the planning results and on the system evaluation, which is the last phase of the complete system planning process. Item-related data are usually collected in a document called the PFEP (Harris, 2003). This document, which is presented as a table, offers information about every item or part needed for production, logistics, and procurement. Typically, the PFEP is an exhaustive document. This means that, for every item, an extensive amount of information is available, most of which is not directly related to the dimensioning phase of the tugger train planning.

The proposed system works in the following way: (a) at the beginning of a production process, the tugger train delivers raw material needed for production, which are packed in different container types (small boxes, medium boxes, big boxes, and spools) according to the PFEP (Table 1), from central warehouse to SM (Figure 4); (b) from the SM, the tugger train delivers each item or part needed for production to the shop floors (S1, S2, S3, S4, and S5 in Figure 4), where raw materials are stored according to the PFEP; (c) when the production process starts, raw materials are used and the number of stored containers decreases; (d) the finished goods are made on the last shop floor, that is, S5; (e) using each raw material container generates a trigger signal, which then calculates raw material replenishment after every trigger when Kanban system is generated; (f) then the calculation shows the number of raw material containers which are required on the shop floor on which the *Kanban trigger* is generated; (g) the raw material containers are prepared and loaded on to the next tugger train; (h) the tugger train leaves the SM and moves along the defined trajectory stopping on each shop floor (S1, S2, S3, S4, and S5), where it is necessary to unload raw material containers; (i) raw material containers are unloaded on the last shop floor S5, and the finished goods are picked up; (j) the tugger train goes to the SM, where finish goods get





unloaded; (k) while the production process is underway, the *Kanban triggers* are generated, and the loading of raw material containers continues in the SM, and the inventory-production process is repeated in cycles from point (e) to point (k).

The characteristics of the production system are as follows: time needed to produce one *final good* is 1 minute, which means 60 pieces are produced per hour. The characteristics of the *Milk run system* are as follows: (a) *average speed* = 4 [km/hr]; (b) *length of route* = 265 [m]; and (c) *time for loading* in SM = 4 [minute].

## 4 | EXPERIMENTAL RESULTS AND DISCUSSION

According to the *Milk run system* characteristics, that is, the *average speed*, *length of route*, *time for loading* in SM, and a *plan for every part*, which was proposed in a previous study (Simić-2, 2019a, 2019b), new optimal characteristics of the *Milk run system* in the automotive industry are shown.

**TABLE 1** Plan for every part (PFEP) and number of pieces in one finished good

| Material description | Bill of Material for one FG | Production needs per hour | Pieces per container | Number of containers per hour | Container type |
|----------------------|-----------------------------|---------------------------|----------------------|-------------------------------|----------------|
| Material 1           | 10                          | 600                       | 2,500                | 0.2400                        | Small box      |
| Material 2           | 10                          | 600                       | 600                  | 1.0000                        | Small box      |
| Material 3           | 2                           | 120                       | 1,000                | 0.1200                        | Small box      |
| Material 4           | 5                           | 300                       | 6,500                | 0.0462                        | Small box      |
| Material 5           | 5                           | 300                       | 5,000                | 0.0600                        | Small box      |
| Material 6           | 1                           | 60                        | 168                  | 0.3571                        | Medium box     |
| Material 7           | 1                           | 60                        | 68                   | 0.8824                        | Medium box     |
| Material 8           | 5                           | 300                       | 1,000                | 0.3000                        | Small box      |
| Material 9           | 32                          | 1920                      | 576                  | 3.3333                        | Small box      |
| Material 10          | 1                           | 60                        | 60                   | 1.0000                        | Medium box     |
| Material 11          | 1                           | 60                        | 52                   | 1.1538                        | Medium box     |
| Material 12          | 1                           | 60                        | 20                   | 3.0000                        | Big box        |
| Material 13          | 1                           | 60                        | 48                   | 1.2500                        | Small box      |
| Material 14          | 1                           | 60                        | 25                   | 2.4000                        | Medium box     |
| Material 15          | 1                           | 60                        | 20                   | 3.0000                        | Big box        |
| Material 16          | 1                           | 60                        | 54                   | 1.1111                        | Small box      |
| Material 17          | 24                          | 1,440                     | 500                  | 2.8800                        | Spool          |
| Material 18          | 20                          | 1,200                     | 1,000                | 1.2000                        | Small box      |
| Material 19          | 20                          | 1,200                     | 1,000                | 1.2000                        | Small box      |

Abbreviation: FG, final good.

**TABLE 2** Train parameters for a train sequence of 20 minutes

| Train sequence 20 (min)    | Needs per hour | Number of containers per trailer | Needed trailers per hour       | Needed trailers per sequence | % of train space |
|----------------------------|----------------|----------------------------------|--------------------------------|------------------------------|------------------|
| Small box                  | 9.86           | 25                               | 0.3944                         | 0.131466667                  | 4%               |
| Medium box                 | 5.79           | 2                                | 2.895                          | 0.965                        | 27%              |
| Big box                    | 6              | 1                                | 6                              | 2                            | 56%              |
| Spool                      | 2.88           | 2                                | 1.44                           | 0.48                         | 13%              |
|                            |                |                                  | 10.7294                        | 3.576466667                  |                  |
| By the 11 trailer per hour |                |                                  | Must have 4 trailers per train |                              |                  |

**TABLE 3** Milk run and Kanban inventory-production system in 30 iterations (Itr.); number of Kanban cards—containers (Kanban #); stock level in shop floor (Stock Level); raw material replenishment (Replen.); cumulative consumption (Cu. Con.), % until Trigger level (Trigger point)

| Material 2 (Kanban #2) |             |          |          | Material 6 (Kanban #2) |             |          |          | Material 9 (Kanban #5) |             |          |          |               |
|------------------------|-------------|----------|----------|------------------------|-------------|----------|----------|------------------------|-------------|----------|----------|---------------|
| Itr.                   | Stock Level | Re-plen. | Cu. Con. | Trigger point          | Stock Level | Re-plen. | Cu. Con. | Trigger point          | Stock Level | Re-plen. | Cu. Con. | Trigger point |
|                        | 1200        |          |          |                        | 336         |          |          |                        | 2880        |          |          |               |
| 1                      | 1000        |          | 200      | 0.33                   | 316         |          | 20       | 0.12                   | 2240        |          | 640      | 1.11          |
| 2                      | 800         |          | 400      | 0.67                   | 296         |          | 40       | 0.24                   | 1600        |          | 1280     | 2.22          |
| 3                      | 600         |          | 600      | 1.00                   | 276         |          | 60       | 0.36                   | 1536        | 576      | 1920     | 3.33          |
| 4                      | 400         |          | 800      | 1.33                   | 256         |          | 80       | 0.48                   | 1472        | 576      | 2560     | 4.44          |
| 5                      | 800         | 600      | 1000     | 1.67                   | 236         |          | 100      | 0.60                   | 1408        | 576      | 3200     | 5.56          |
| 6                      | 600         |          | 1200     | 2.00                   | 216         |          | 120      | 0.71                   | 1344        | 576      | 3840     | 6.67          |
| 7                      | 400         |          | 1400     | 2.33                   | 196         |          | 140      | 0.83                   | 1280        | 576      | 4480     | 7.78          |
| 8                      | 800         | 600      | 1600     | 2.67                   | 176         |          | 160      | 0.95                   | 1216        | 576      | 5120     | 8.89          |
| 9                      | 600         |          | 1800     | 3.00                   | 156         |          | 180      | 1.07                   | 1152        | 576      | 5760     | 10.00         |
| 10                     | 400         |          | 2000     | 3.33                   | 136         |          | 200      | 1.19                   | 1088        | 576      | 6400     | 11.11         |
| 11                     | 800         | 600      | 2200     | 3.67                   | 284         | 168      | 220      | 1.31                   | 1600        | 1152     | 7040     | 12.22         |
| 12                     | 600         |          | 2400     | 4.00                   | 264         |          | 240      | 1.43                   | 1536        | 576      | 7680     | 13.33         |
| 13                     | 400         |          | 2600     | 4.33                   | 244         |          | 260      | 1.55                   | 1472        | 576      | 8320     | 14.44         |
| 14                     | 800         | 600      | 2800     | 4.67                   | 224         |          | 280      | 1.67                   | 1408        | 576      | 8960     | 15.56         |
| 15                     | 600         |          | 3000     | 5.00                   | 204         |          | 300      | 1.79                   | 1344        | 576      | 9600     | 16.67         |
| 16                     | 400         |          | 3200     | 5.33                   | 184         |          | 320      | 1.90                   | 1280        | 576      | 10240    | 17.78         |
| 17                     | 800         | 600      | 3400     | 5.67                   | 164         |          | 340      | 2.02                   | 1216        | 576      | 10880    | 18.89         |
| 18                     | 600         |          | 3600     | 6.00                   | 144         |          | 360      | 2.14                   | 1152        | 576      | 11520    | 20.00         |
| 19                     | 400         |          | 3800     | 6.33                   | 292         | 168      | 380      | 2.26                   | 1088        | 576      | 12160    | 21.11         |
| 20                     | 800         | 600      | 4000     | 6.67                   | 272         |          | 400      | 2.38                   | 1600        | 1152     | 12800    | 22.22         |
| 21                     | 600         |          | 4200     | 7.00                   | 252         |          | 420      | 2.50                   | 1536        | 576      | 13440    | 23.33         |
| 22                     | 400         |          | 4400     | 7.33                   | 232         |          | 440      | 2.62                   | 1472        | 576      | 14080    | 24.44         |
| 23                     | 800         | 600      | 4600     | 7.67                   | 212         |          | 460      | 2.74                   | 1408        | 576      | 14720    | 25.56         |
| 24                     | 600         |          | 4800     | 8.00                   | 192         |          | 480      | 2.86                   | 1344        | 576      | 15360    | 26.67         |
| 25                     | 400         |          | 5000     | 8.33                   | 172         |          | 500      | 2.98                   | 1280        | 576      | 16000    | 27.78         |
| 26                     | 800         | 600      | 5200     | 8.67                   | 152         |          | 520      | 3.10                   | 1216        | 576      | 16640    | 28.89         |
| 27                     | 600         |          | 5400     | 9.00                   | 132         |          | 540      | 3.21                   | 1152        | 576      | 17280    | 30.00         |
| 28                     | 400         |          | 5600     | 9.33                   | 280         | 168      | 560      | 3.33                   | 1088        | 576      | 17920    | 31.11         |
| 29                     | 800         | 600      | 5800     | 9.67                   | 260         |          | 580      | 3.45                   | 1600        | 1152     | 18560    | 32.22         |
| 30                     | 600         |          | 6000     | 10.00                  | 240         |          | 600      | 3.57                   | 1536        | 576      | 19200    | 33.33         |

#### 4.1 | Experimental results for the Milk run system

The goal of this study is to realize an optimal Milk run tact per train (raw material replenishment frequency) of 20 [minute]; therefore, the total numbers of trains per hour is defined as 3 (route frequency). Table 2 shows the Milk run calculation for train parameters for a raw material replenishment frequency (train sequence) of 20 minutes, and it presents the average value calculated for 1 hr. The value, 3.57, suggests that four trailers per train are necessary for a continuous production process.

Recent research extends the existing Milk run system in combination with the Kanban system as a raw material trigger to the inventory-production system in the automotive industry. As mentioned before, the Milk run system presents only the average value calculated for 1 hr, but recent research calculates the raw material replenishment after every trigger when the Kanban system is generated.

## 4.2 | Experimental results for the Milk run and Kanban systems

According to the nature of PSO, the experiment is repeated 100 times with the data collection and experimental results to optimize the inventory-production system for minimizing the total available capacity. In particular, (a) the inventory capacity in super market, (b) the capacity in tigger train transportation, and (c) material stock level on shop floor should be minimized; furthermore, a smooth inventory-production process should be achieved.

In this research, the term *iteration* presents the *Milk Run* tact per train in 20 minutes. The *Inventory analysis* for 30 *Iterations* for only three raw materials, Material 2, Material 6, and Material 9, is presented in Table 3.

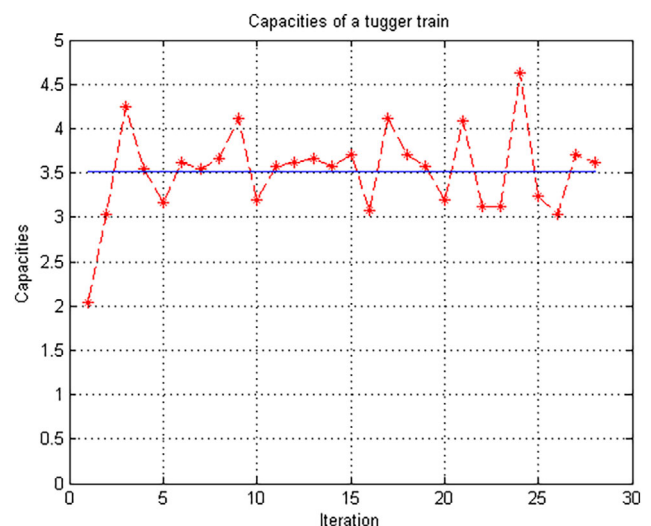
Table 3 presents the *Milk run* and Kanban inventory-production system in 30 iterations. It should be noted that there are different numbers of Kanban cards for different products, thereby indicating that, at the beginning of a production process, there are different numbers of containers on different shop floors. The table presents the *number of Kanban cards* (containers), *stock level in shop floor*; *raw material replenishment*; and *cumulative consumption* until the *trigger level*. The *trigger level* after every iteration increases, for every raw material, and when it reaches the *Trigger point*, the Kanban system generates information that the existing shop floor needs raw material replenishment—new container(s)—which is marked as YELLOW in Table 3.

After every iteration, usually, some *Trigger points* are generated. At that moment, the workers in the SM start to prepare the needed raw material containers. One iteration is necessary for preparation of the raw material containers, and the next iteration is used for the delivery of the raw material containers. After the *Kanban trigger point* is reached, two iterations are necessary for delivering the raw material replenishment container to the Kanban shop floor. This point is marked GREEN in Table 3. The entire inventory-production process continues from *Trigger point*, that is, SM preparation of raw material containers, loading of the raw material containers on tigger train trailers in the SM, transportation of the raw materials, delivery of the containers to the Kanban shop floor, which generates the *Kanban Trigger point*, and picking-up of finished goods from the last shop floor.

The capacities of a trailer and loading of the trailer in a tigger train after the Kanban trigger point is reached after the third, thirteenth, twenty-third, and twenty-sixth iteration are presented in Tables 4–7. Table 4 presents the experimental results of used capacities in the tigger train in the third iteration. It is also presents the number of iterations (*iteration*), number of pieces (*pie.*), maximum number of pieces per trail for different container types (*ma.*), number of pieces in trailers from one (I) to five (V) in one tigger train, and the capacities for each trailer and for the entire tigger train.

**TABLE 4** Capacity of a trailer of the tigger train after three iterations

| iteration | pie. | ma. | trailers |      |      |      |      |
|-----------|------|-----|----------|------|------|------|------|
|           |      |     | I        | II   | III  | IV   | V    |
| 3         |      |     |          |      |      |      |      |
| Small box | 1    | 25  |          |      | 1    |      |      |
| Medium b. | 0    | 2   |          |      |      |      |      |
| Big box   | 2    | 1   | 1        | 1    |      |      |      |
| Spool     |      | 2   |          |      |      |      |      |
|           | 3    |     | 1.00     | 1.00 | 0.04 | 0.00 | 0.00 |
|           |      |     |          |      |      |      | 2.04 |



**FIGURE 5** Capacities of a tigger train after the Kanban trigger; loading the containers in the trailers for the first 30 iterations

It is important to notice that the used capacities in the tugger train go from a minimum of 2.04 to a maximum of 4.62. The average value is 3.51, and the  $SD$  is  $\pm 0.49$ . The used capacities in the tugger train are presented in Figure 5. This means that the minimum capacity of 2.04 translates into three trailers, and the maximum capacity of 4.62 translates into five trailers, which indicates that the used tugger train transportation system has five trailers. The characteristic values of the capacities of a trailer and loading the trailer of a tugger train, after the Kanban trigger, are presented in Tables 4–7 and in Figure 5.

It is interesting to compare the existing *Milk run* production-delivery system presented in (Simić-2, 2019a, 2019b) and experimental results of recent research where *Milk run* is extended with the Kanban system. On one hand, the *Milk run* production-delivery system shows a value of 3.57, which translates into four trailers per train, without PSO optimization. On the other hand, the *Milk run* and Kanban system supported with PSO optimization present a trailer capacity of  $3.51 \pm 0.49$ , which also translates into four trailers per train. It could be concluded that, with different initial values and some constraints, it is probably possible to optimize this inventory-production process with four trailers per train.

However, this opens a new chapter of research, “How to Calculate the Initial Value of Iteration” for this optimization and which new constraints could be added to the inventory-production process. The problem regarding the initial values can be solved as follows: (a) number of Kanban cards—containers at the beginning of a production process, (b) but it opens discussion of inventory space, number of shelves, (c) and design of every shop floor, (d) and/or usage of different means for industrial material transportation in inbound logistics. All these ideas open new fields of future research.

### 4.3 | Benefits and limitations of the Milk run and Kanban systems

Companies that implement *Milk run* and Kanban systems as a concept of internal material flow reap various benefits. This system can be implemented in every serial production environment such as those in the electrical industry, electronic industry, mechanical industry. It helps to

| iteration<br>13 | pie. | ma. | trailers |      |      |      |      |
|-----------------|------|-----|----------|------|------|------|------|
|                 |      |     | I        | II   | III  | IV   | V    |
| Small box       | 2    | 25  |          |      |      | 2    |      |
| Medium b.       | 2    | 2   |          |      | 2    |      |      |
| Big box         | 2    | 1   | 1        | 1    |      |      |      |
| Spool           | 1    | 2   |          |      |      | 1    |      |
|                 | 7    |     | 1.00     | 1.00 | 1.00 | 0.58 | 0.00 |
|                 |      |     |          |      |      |      | 3.58 |

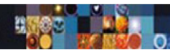
**TABLE 5** Capacity of a trailer of the tugger train after 30 iterations

| iteration<br>23 | pie. | ma. | trailers |      |      |      |      |
|-----------------|------|-----|----------|------|------|------|------|
|                 |      |     | I        | II   | III  | IV   | V    |
| Small box       | 2    | 25  |          |      |      |      | 2    |
| Medium b.       | 3    | 2   |          |      | 2    | 1    |      |
| Big box         | 2    | 1   | 1        | 1    |      |      |      |
| Spool           | 1    | 2   |          |      |      | 1    |      |
|                 | 8    |     | 1.00     | 1.00 | 1.00 | 1.00 | 0.08 |
|                 |      |     |          |      |      |      | 4.08 |

**TABLE 6** Capacity of a trailer of the tugger train after 23 iterations

| iteration<br>26 | pie. | ma. | trailers |      |      |      |      |
|-----------------|------|-----|----------|------|------|------|------|
|                 |      |     | I        | II   | III  | IV   | V    |
| Small box       | 3    | 25  |          |      |      |      | 3    |
| Medium b.       | 4    | 2   |          |      | 2    | 2    |      |
| Big box         | 2    | 1   | 1        | 1    |      |      |      |
| Spool           | 1    | 2   |          |      |      |      | 1    |
|                 | 10   |     | 1.00     | 1.00 | 1.00 | 1.00 | 0.62 |
|                 |      |     |          |      |      |      | 4.62 |

**TABLE 7** Capacity of a trailer of the tugger train after 26 iterations



reduce or eliminate many different types of waste in accordance with the principles of lean production. The *Milk run* optimizes transport capacity usage because of the transportation of both full and empty containers containing raw materials and finished goods. It assures continuous production, eliminating production stoppages due to missing material. Additionally, the *Milk run* reduces the material handling space in production; thus, there is more free space for production activities.

The *Milk run* system is combined with the Kanban system because the Kanban system highlights when something is needed in the production line, and the *Milk run* assures that it is transported at the right time and in an optimal way. On the other hand, these systems are suitable for extremely big production companies. In general, good implementation takes a relatively long time to achieve. Practice has shown that good implementation of the system can be achieved in approximately 2–3 years.

The implementation of the *Milk run* and Kanban systems could generate positive economic benefits for companies because of material flow optimization. Material lots could become smaller, and there would be a smaller quantity of material in circulation, which translates into significant inventory reduction. Furthermore, optimized transport requires less human and transport resources. It is important to note that the inventory is a burden to a company because inventory is trapped capital and it is therefore desirable for inventory levels to be as low as possible.

## 5 | CONCLUSION AND FUTURE WORK

A biological swarm intelligence model comprising particle swarm optimization for modelling material flow optimization using *Milk run* and Kanban systems in automotive industry is presented in this paper. The proposed models optimize the inventory-production system for minimizing the total available capacity by (a) minimizing inventory capacity in the supermarket, (b) minimizing capacity in tugger train transport, and (c) minimizing material stock level on shop floor. At the same time, a smooth inventory-production process is achieved.

The new modified PSO model for the inventory-production system in a unique optimization model is presented in this paper. The major modification to the original PSO is using the capacity of a container instead of a velocity component. For each Kanban trigger, the total timing for the *Milk run* delivery for the necessary raw materials, capacity of every shop floor, and capacity of the trailers of the tugger train are calculated. In the proposed system, the capacity of the container and the quantity of material in circulation are decreased, which presents significant inventory reduction.

Experimental results encourage further research. On one hand, the proposed system could be improved by using soft computing techniques, which could stabilize deviation of the number of trails, or changing the number of Kanban cards (containers at the beginning of a production process) and improving the inventory-production process. The optimization method, PSO, has several parameters that determine its behaviour and efficacy. Future studies could explore how to calculate the initial value of iteration for this optimization model and which new constraints could be added in this specific production process. Future work could focus on extending research on good choice of parameters to various optimization scenarios, which should help production managers to achieve different solutions and make better operational solutions and achieve production results with little effort.

## CONFLICT OF INTEREST

None.

## ORCID

Dragan Simić  <https://orcid.org/0000-0001-7392-9767>

## REFERENCES

- Aghajani, M., Keramati, A., & Javadi, B. (2012). Determination of number of kanban in a cellular manufacturing system with considering rework process. *The International Journal of Advanced Manufacturing Technology*, 63, 1177–1189. <https://doi.org/10.1007/s00170-012-3973-y>
- Aksoy, A., & Öztürk, N. (2016). A two-stage method to optimize the milk-run system. *European Journal of Engineering Research and Science*, 1, 7–11.
- Chee, S. L., Chong, M. Y., & Chin, J. F. (2012). Milk-run kanban system for raw printed circuit board withdrawal to surface-mounted equipment. *Journal of Industrial Engineering and Management*, 5, 382–405. <https://doi.org/10.3926/jiem.352>
- Chen, L., & Meng, B. (2010). Application of value stream mapping based lean production system. *International Journal of Business and Management*, 5, 203–209.
- Duri, D., Frein, Y., & Di Mascio, M. (2000). Comparison among three pull control policies: Kanban, base stock, and generalized kanban. *Annals of Operations Research*, 93, 41–69. <https://doi.org/10.1023/A:1018919806139>
- Emde, S., & Gendreau, M. (2017). Scheduling in-house transport vehicles to feed parts to automotive assembly lines. *European Journal of Operational Research*, 260, 255–267. <https://doi.org/10.1016/j.ejor.2016.12.012>
- Hardi, H. P., Fitra, A., & Nindiani, A. (2019). Control and integration of milk-run operation in Japanese automotive company in Indonesia. *Management and Production Engineering Review*, 10, 79–88. doi:10.24425/mper.2019.128246
- Harris, R., Harris, C., & Wilson, E. (2003). *Making materials flow*. Boston, MA, USA: Lean Enterprise Institute.
- Herer, Y. T., & Shalom, L. (2000). The Kanban assignment problem—A non-integral approach. *European Journal of Operational Research*, 120, 260–276. [https://doi.org/10.1016/S0377-2217\(99\)00155-1](https://doi.org/10.1016/S0377-2217(99)00155-1)



- Ilin, V., Ivetić, J., & Simić, D. (2017). Understanding the determinants of e-business adoption in ERP-enabled firms and non-ERP-enabled firms: A case study of the Western Balkan Peninsula. *Technological Forecasting and Social Change*, 125, 206–223. <https://doi.org/10.1016/j.techfore.2017.07.025>
- Kennedy, J., & Eberhart, R. (1995). Particle swarm optimization. *IEEE International Conference on Neural Networks*, 4, 1942–1948. <https://doi.org/10.1109/ICNN.1995.488968>
- Kodym, O., Čujan, Z., Turek, M., & Mikušová, N. (2019). Optimization of material flow using simulation. *MATEC Web of Conferences*, 263. <https://doi.org/10.1051/mateconf/201926301007>
- Krieg, G. (2005). *Kanban-controlled manufacturing systems*. Berlin, Heidelberg: Springer-Verlag.
- Krolczyk, J. B., Krolczyk, G. M., Legutko, S., Napiorkowski, J., Hloch, S., Foltys, J., & Tama, E. (2015). Material flow optimization—A case study in automotive industry. *Tehnicki Vjesnik—Technical Gazette*, 22, 1447–1456. doi:10.17559/TV-20141114195649
- Kundua, K., Rossini, M., & Portioli-Staudacher, A. (2019). A study of a kanban based assembly line feeding system through integration of simulation and particle swarm optimization. *International Journal of Industrial Engineering Computations*, 10, 421–442. <https://doi.org/10.5267/j.ijiec.2018.12.001>
- Mao, Z., Huang, D., Fang, K., Wang, C., & Lu, D. (2019). Milk-run routing problem with progress-lane in the collection of automobile parts. *Annals of Operations Research*. <https://doi.org/10.1007/s10479-019-03218-x>
- Patel, D. R. (2017). *Design and optimization of Milk-run material supply system with simultaneous pickups and deliveries in time windows*. (Ph.D. thesis). Gujarat Technological University, Ahmedabad, India.
- Pinheiro, D. A., Jr., Galvão, A. N. N., & Mendes, M. M. L. (2019). An agent-based approach to evaluate collaborative strategies in milk-run OEM operations. *Computers & Industrial Engineering*, 129, 545–555. <https://doi.org/10.1016/j.cie.2019.01.026>
- Sarker, B. R., & Balan, C. V. (1998). Operations planning for a single-stage kanban system operating under linear demand. *International Journal of Production Research*, 36, 357–375. <https://doi.org/10.1080/002075498193787>
- Simić, D., & Simić, S. (2012). Hybrid artificial intelligence approaches on vehicle routing problem in logistics distribution. In *Hybrid artificial intelligence systems, lecture notes in computer science* (Vol. 7208, pp. 208–220). Berlin, Heidelberg: Springer. [https://doi.org/10.1007/978-3-642-28942-2\\_19](https://doi.org/10.1007/978-3-642-28942-2_19)
- Simić, D., Kovačević, I., Svirčević, V., & Simić, S. (2015). Hybrid firefly model in routing heterogeneous fleet of vehicles in logistics distribution. *Logic Journal of the IGPL*, 23, 521–532. <https://doi.org/10.1093/jigpal/jzv011>
- Simić, D., Svirčević, V., Ilin, V., Simić, S. D., & Simić, S. (2019a). Particle swarm optimization and pure adaptive search in finish goods' inventory management. *Cybernetics and Systems*, 50, 58–77. <https://doi.org/10.1080/01969722.2018.1558014>
- Simić, D., Svirčević, V., Ilin, V., Simić, S. D., & Simić, S. (2019b). Material flow optimization using milk run system in automotive industry. In *Advances in intelligent systems and computing* (Vol. 950, pp. 411–421). Cham, Switzerland: Springer. [https://doi.org/10.1007/978-3-030-20055-8\\_39](https://doi.org/10.1007/978-3-030-20055-8_39)
- Sitek, P., & Wikarek, J. (2019). Capacitated vehicle routing problem with pick-up and alternative delivery (CVRPPAD): Model and implementation using hybrid approach. *Annals of Operations Research*, 273, 257–277. <https://doi.org/10.1007/s10479-017-2722-x>
- Song, D. P. (2013). *Optimal control and optimization of stochastic supply chain systems*. London, England: Springer-Verlag.
- Takahashi, K., & Myreshka, & Hirotani, D. (2005). Comparing CONWIP, synchronized CONWIP and kanban in complex supply chains. *International Journal of Production Economics*, 93–94, 25–40. <https://doi.org/10.1016/j.ijpe.2004.06.003>
- Urru, A., Bonini, M., & Echelmeyer, W. (2018). Planning of a milk-run systems in high constrained industrial scenarios. In *Proceeding of 22nd IEEE international conference on intelligent engineering systems* (pp. 231–238). Las Palmas de Gran Canaria, Spain: IEEE. <https://doi.org/10.1109/INES.2018.8523926>
- Wagner, B., &ENZLER, S. (2006). *Material flow management*. Heidelberg, Germany: Physica-Verlag.
- Widyadana, G. A., Wee, H. M., & Chang, J.-Y. (2010). Determining the optimal number of kanban in multi-products supply chain system. *International Journal of Systems Science*, 41, 189–201. <https://doi.org/10.1080/00207720903042996>
- Wu, Q., Wang, X., He, Y. D., Xuan, J., & He, W. D. (2018). A robust hybrid heuristic algorithm to solve multi-plant milk-run pickup problem with uncertain demand in automobile parts industry. *Advances in Production Engineering & Management*, 13, 169–178. <https://doi.org/10.14743/apem2018.2.282>

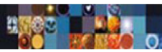
## AUTHOR BIOGRAPHIES

**Dragan Simić** received the Ph.D. degree at the Department of Mathematics and Computer Sciences, Faculty of Sciences, University of Novi Sad, Novi Sad, Serbia. More than 25 years he worked in industry. He is currently a Full Professor at the Department of Transport Engineering, Faculty of Technical Sciences, University of Novi Sad, Novi Sad, Serbia. His current research interest includes: information systems, decision support systems, hybrid artificial intelligence systems, soft computing, and their applications in logistics, production, financial forecasting, and medicine. He is a member of the International Federation for Information Processing (IFIP) – and represents Republic of Serbia in Council and Technical Committee TC 8 – Information Systems from 2007.

**Vasa Svirčević** received the Ph.D. degree at the Department of Transport Engineering, Faculty of Technical Sciences, University of Novi Sad, Novi Sad, Serbia. More than 20 years he worked in industry as purchase manager and logistics manager in multinational companies: Lafarge (France), Philip Morris (United States), Lear Corporation (United States). He is currently a freelancer and consultant in some multinational companies.

**Emilio Corchado** received the Ph.D. degree in computer science at the Department of Computer Science and Automatics, University of Salamanca, Salamanca, Spain, where he is currently a Full Professor in computer and automatic science. He has published over 100 peer-reviewed articles in a range of topics from knowledge management and risk analysis, intrusion detection systems, food industry, artificial vision, and the modeling of industrial processes. His research interests include neural networks, with a particular focus on exploratory projection pursuit, maximum likelihood hebbian learning, self-organizing maps, multiple classifier systems, and hybrid systems. He is an RTD Expert hired by





international organizations such as the European Commission, the Grant Agency of Czech Republic, the Spanish National Agency for Assessment and Forecasting, and he has collaborated with SMEs and new companies in the innovation field in about 40 projects. He has patented software models, and more than 10 ICT tools and models. He has been the Organizing Chair, the Program Committee Chair, the Session Chair, and the General Chair of a number of conferences. He was the Chair of the IEEE Spanish Section, from 2014 to 2015, and he has actively contributed to several current projects in the EU including SOFTCOMP, IT4Innovation, ICT Action COST IC1303, and IntelliCIS NISIS.

**José Luis Calvo Rolle** received M.S. and Ph.D. degrees in Industrial Engineering from the University of Leon in 2004 and 2007, respectively. He is an Associate Professor of Automatic Control and the head of the Industrial Engineering Department, Faculty of Engineering, University of A Coruña, Spain. His main research areas are associated with the application of expert system technologies in diagnosis and control systems and in intelligent training systems for control engineering.

**Svetislav D. Simić** is currently an Associate Teaching at the Department of Computer Sciences and Informatics, Faculty of Technical Sciences, University of Novi Sad, Novi Sad, Serbia. He finished Engineering Bachelor with Honours in Computing and Control Engineering – subfield Applied Computer Science and Informatics (GPA 10.00/10.00). His mainly research areas are focused on the field of artificial intelligence and applications in intelligent systems.

**Svetlana Simić** received his PhD at the Faculty of Medicine, University of Novi Sad, Serbia. She is Associated Professor of Neurology at Faculty of Medicine, University of Novi Sad and neurologist at Clinic of neurology, Clinical Center of Vojvodina in Serbia. Her main scientific interest is related to headache and pain. Because of that she attended training such as: EFIC's School for: Neurological diagnosis in chronic pain – Montescano (Italy), Tempus Palliative and Pain Medicine Project – Florence (Italy) and Lyon (France), International Teaching Course in Neurology – Targu-Mures (Romania), ECPD International Postgraduate Studies: Basic, Advanced and Higher course in Acupuncture. She is founder and Vice president of Serbian Headache Society.

**How to cite this article:** Simić D, Svirčević V, Corchado E, Calvo-Rolle JL, Simić SD, Simić S. Modelling material flow using the Milk run and Kanban systems in the automotive industry. *Expert Systems*. 2021;38:e12546. <https://doi.org/10.1111/exsy.12546>